

Biswash Ghimire

Contact:

(517) 899-1751 | ghi.biswash@gmail.com
[LinkedIn](#) | biswashghi.github.io

Summary

Lead Software Engineer with 5+ years designing distributed backend systems, event-driven platforms, and reusable cloud-native services for large-scale financial applications. Technical lead experienced in cross-team architecture, platform modernization, and partner integrations.

Experience

Capital One / Lead Software Engineer

Aug 2020 - Present, Remote

Loyalty Earn Platform & Distributed Systems

- Technical lead within a 30-40 engineer, six-team modernization effort building Capital One's next-generation Loyalty Earn Platform.
- Architected event-driven processing supporting 60M+ daily events, reducing rewards latency by 8 hours and processing runtime by 3 hours.
- Designed multi-region event-processing architectures incorporating layered idempotency, event ordering, and consistency strategies to ensure reliable reward processing across distributed streaming systems.
- Redesigned reward processing architecture to support ordered benefit evaluation and threshold-based state transitions within an event-driven platform to support Discover integration, overcoming assumptions of benefit independence.

Partner Integrations & Platform Engineering

- Designed the convergence strategy from legacy orchestration APIs to a unified Loyalty Platform service supporting Amazon, PayPal, REI, T-Mobile, and future partners.
- Consolidated 10 partner API endpoints into a common integration platform processing 30M+ daily API requests, using reusable abstractions to simplify onboarding of new partners.
- Balanced repository architecture and deployment tradeoffs by converging partner implementations into a shared codebase with strict automated regression guarantees, enabling independent feature development while maintaining partner-specific behavior.
- Modernized AutoRedemption through reusable event-driven integration patterns that became the architectural foundation for Discover AutoRedemption.
- Generalized bureau data pipelines supporting Equifax, Experian, and TransUnion, enabling reusable partner onboarding with 100% migration validation.

Platform Modernization & Technical Leadership

- Technical lead for a team of five engineers, partnering with engineering leadership and product to define technical direction, architecture, and implementation plans.
- Served as a Loyalty platform SME, collaborating with peer tech leads and Distinguished Engineers on event schemas, integration contracts, and cross-team architecture.
- Regularly collaborated with engineering managers, product owners, and peer technical leads to decompose large architectural initiatives into implementation roadmaps spanning multiple engineering teams.

University of Michigan EECS Department / Instructional Aide

Jan 2019 - Apr 2020

- Led lab sessions, held office hours, and supported student understanding of web technologies such as REST APIs, CSR/SSR, DNS, and cloud computing.
- Collaborated with faculty to design, administer, and grade exams and assignments; provided detailed feedback to students and contributed to continuous course improvement through content development and updates.

Scientific Computing and Flow Physics Laboratory / Research Assistant

Sep 2016 - Apr 2017

- Worked under a PhD candidate to explore benefits of high-order vs low-order discretization in space and time when utilizing numerical methods such as Runge-Kutta (RK), Discontinuous-Galerkin (DG), and Finite Difference (FD) for unsteady physical system simulation using GMSH.
- Presented learnings in a poster presentation session at the end of the year organized by the

Projects	Novel Tracker – Cross-Platform Browser Extension Personal project • Built a local-first Manifest V3 extension shipping to Chrome, Edge, Firefox (desktop and Android), and a native Safari Web Extension for macOS/iOS/iPadOS, auto-tracking reading progress across web-novel sites via 9 site-specific parsers with a generic-metadata fallback. • Designed an optional cloud-sync layer (Node.js API, Google sign-in) using Hybrid Logical Clocks and remove-wins merge rules to deterministically resolve offline edit conflicts across devices. • Unified storage, parsing, and sync logic into shared modules reused across all four browser targets; automated release packaging and end-to-end testing (Playwright) via GitHub Actions.
Skills	Languages: Go, Python, Java, C, C++ Frameworks & Tools: Spring, Docker, Spark, Hadoop Databases: PostgreSQL, Cassandra, MongoDB, DocumentDB, DynamoDB Cloud Platforms / Services: AWS (Lambda, EC2, EMR, Fargate, SQS, SNS, Kinesis, and more)
Education	University of Michigan / Computer Science, B.S. Aug 2016 - May 2020, Ann Arbor, MI • Thread Manager: Developed a multi-threaded library for Unix operating systems in C++, supporting threading primitives, mutexes, and more. • Key/Value Store: Implemented a Go-based key-value service supporting concurrent datastore requests via primary/backup replication and a Paxos-based replicated state machine.
Certifications	AWS Solutions Architect May 2021 - Present